



East West University
Department of Computer Science and Engineering

Course: CSE 475 (Machine Learning)
Section - 03

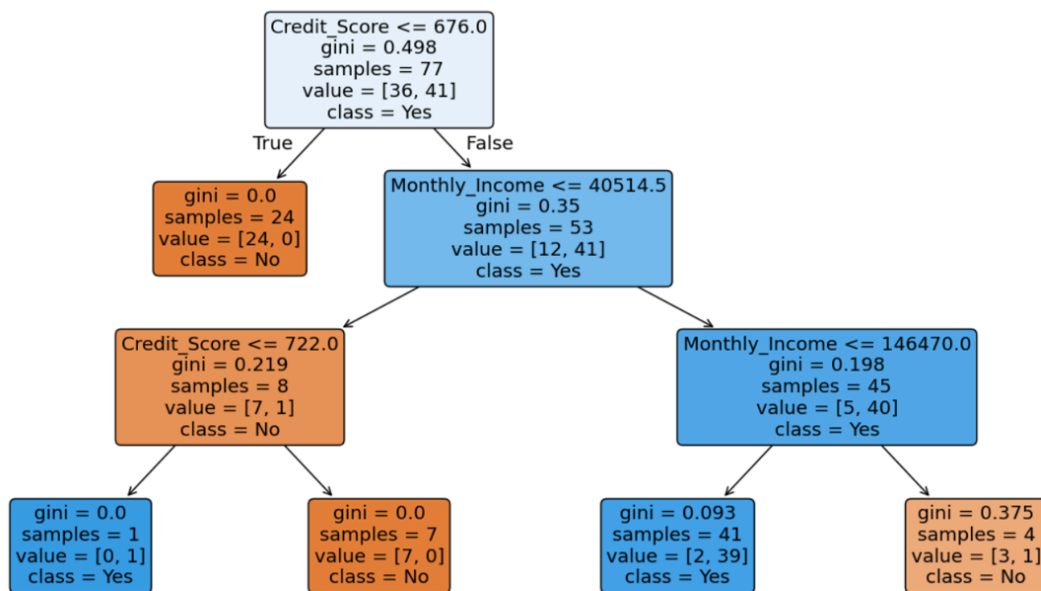
Submitted To :	Submitted By :
Dr. Md. Tauhid Bin Iqbal Assistant Professor Department of CSE East West University	Name : Hafsa Ferdousi Student ID : 2023-3-60-321 Department : CSE Date : 06 /09 / 2026

Lab Report:02

K-NN Classification

K Value	Distance Metric	90:10	80:20	70:30
3	Euclidean	50%	60%	63%
3	Manhattan	40%	50%	57%
5	Euclidean	50%	65%	60%
5	Manhattan	60%	70%	60%
7	Euclidean	50%	60%	60%
7	Manhattan	50%	65%	63%

Decision Tree (CART)



	Criterion	Max Depth	Split	Accuracy (%)	Log Loss
0	gini	3	90:10	80.00	7.3041
1	gini	5	90:10	80.00	7.3304
2	gini	7	90:10	80.00	7.3304
3	gini	3	80:20	90.00	3.6625
4	gini	5	80:20	90.00	3.6806
5	gini	7	80:20	85.00	3.6998
6	gini	3	70:30	93.33	2.4502
7	gini	5	70:30	93.33	2.4537
8	gini	7	70:30	90.00	2.4666
9	entropy	3	90:10	50.00	7.5176
10	entropy	5	90:10	60.00	14.4734
11	entropy	7	90:10	60.00	14.4580
12	entropy	3	80:20	75.00	3.7896
13	entropy	5	80:20	75.00	7.2839
14	entropy	7	80:20	75.00	7.2839
15	entropy	3	70:30	83.33	2.5264
16	entropy	5	70:30	83.33	4.8560
17	entropy	7	70:30	83.33	4.8560

Link:

 KNN.ipynb

 Decision Tree (CART).ipynb